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A coupled thermo-mechanical model based on the combined finite-discrete element method for simulating thermal cracking of rock

Chengzeng Yan[a], Hong Zheng[b,*]

[a] Faculty of Engineering, China University of Geosciences, Wuhan 430074, Hubei, China
[b] State Key Laboratory of Geomechanics and Geotechnical Engineering, Institute of Rock and Soil Mechanics, Chinese Academy of Sciences, Wuhan, Hubei, China

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ABSTRACT

Based on a combined finite-discrete element method (FDEM), this study builds a coupled thermo-mechanical model (termed FDEM-TM) to simulate thermal cracking of rock. The coupled thermo-mechanical model consists of two major parts. In the first part the temperature distribution of the system is analyzed based on the heat conduction equation. In the second part the thermal stress caused by temperature change is added to perform mechanical fracture calculation. Based on these two parts, we can model rock fracture driven by thermo-mechanical coupling. Three examples with analytic solutions are used to verify the correctness of the model in dealing with the problems of steady-state heat conduction, unsteady-state conduction, and thermal-mechanic coupling, respectively. In addition, an example of thermal cracking is also given and compared with the experimental results. The simulation results are in excellent agreement with the analytical solutions or experimental results, verifying the correctness of the coupled thermo-mechanical model to simulate thermal cracking. The proposed method provides a new tool for thermal-mechanical coupling problems in geothermal exploitation.

1. Introduction

Rock is composed of different mineral grains or crystals cemented together. The thermal expansion characteristics of different mineral grains are different. Consequently, even in the case of uniform heating, thermal stress will be produce in rock. Once the stress exceeds the strength of the material, rock will generate micro-cracks. If these micro-cracks constitute fracture networks, macroscopic thermal cracking will emerge. From geothermal exploitation, nuclear waste storage, oil exploitation, coal seam gas safety drainage and crustal evolution, to earthquake mechanism research, we need to consider rock thermal cracking. For example, in a nuclear waste repository, we should prevent rock thermal cracking induced by decay heat of nuclear waste, because rock fracture will cause radionuclide migration contamination. However, in geothermal extraction and petroleum exploitation, thermal cracking can be utilized to form fracture network in rock, which will enhance the permeability of rock, thereby increasing oil production and geothermal acquisition amount. Consequently, the study of thermal cracking has important theoretical and engineering significance.

Many scholars have been studied thermal cracking and the effect of temperature on mechanical properties of rock though experiment.[1,2] Another mean of studying thermal cracking of rocks is numerical simulation. The main numerical methods for simulating thermal cracking can be divided into two categories: continuum-based methods and discontinuum-based methods.

For continuum-based methods, Tang et al.[3,4] established a coupled thermo-mechanical model in RFPA to simulate thermal cracking process of brittle materials. Similarly, Li et al.[5,6] built an initially thermo-mechanical-damage model (TMD) and a temperature-seepage-stress-damage (THMD) model to explore meso-structural damage and evolutionary mechanisms of rock under the effect of multi-physics coupling. Using Flac3D, Kwon and Cho [7] studied the effect of excavation damage zone on rock behavior of thermo-mechanical coupling and hydraulic-mechanical coupling, but it cannot simulate thermal cracking of rock. Ngo et al.[8] proposed a new thermal damage model based on finite element method, it is possible to model softening behavior of brittle materials under the effect of load, including temperature change. Dempsey et al.[9] studied the stress change and permeability enhancement caused by fluid injection in geothermal systems with thermo-hydraulic-mechanical simulation program (FEHM).

* Corresponding author.
E-mail addresses: yanchengzheng86@gmail.com (C. Yan), hzheng@whrsm.ac.cn (H. Zheng).

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For discontinuum-based methods, using discontinuous deformation analysis (DDA), Jiao et al.[10] proposed a coupled thermo-mechanical model that can simulate the rock fracture induced by temperature stress. Wanne and Young[11] performed a numerical simulation of rock thermal cracking based on BPM model. Feng et al.[12,13] proposed a novel discrete thermal element method (DTEM) for modeling heat transfer in systems comprising a large number of circular particles. Xia and Zhao[14,15] presented a new thermo-mechanical coupled particle model to simulate rock damage caused by thermal stress. The difference between the model and the traditional thermo-mechanical particle model is that the elastic modulus and strength parameters are defined as a function of temperature explicitly. Using the particle flow code, Dedecker et al.[16] studied the effect of high pressure fluid and thermal stress on rock mechanical behavior. Ali and Bradshaw[17,18] studied the effect of microwave heating on the mineral crushing using BPM model in particle flow code. Vargas-Escobar[19] proposed a particle thermal dynamics (TPD) for simulating heat transfer in particle material.

Although the above numerical methods have achieved some good results in the simulation of thermal cracking, they did not consider contacts explicitly, or limit to solve small-scale problems. To simulate continuum fracture, Munjiza et al.[20–22] proposed a combined finite-discrete element method (FDEM), which is very suitable to simulate rock fracture. Since stress and strain in the triangular element are calculated based on finite element method, the concept of stress and strain in continuum mechanics are well preserved in FDEM. Also, in order to reflect discontinuities easily, discrete element method is used to handle contacts in FDEM. In addition, explicit algorithm is used, which enables FDEM to solve large-scale problems. Because of the obvious advantages of this method, FDEM has gained rapid development and extensive application in recent years. For example , a FDEM software package called the Hybrid Optimization Software Suite (HOSS) is developed by Rougier and Knight,[23,24] which includes a large strain finite element formulation addressed in detail in Munjiza et al.[25] Lei et al.[26] developed a generated anisotropic deformation formulation for geomaterials. Mahabadi et al.[27] performed three-dimesnsional FDEM numerical simulation of failure processes observed in Opalinus Clay laboratory samples. Lisjak et al.[28] studied the failure mechanisms around unsupported circular excavations in anisotropic clay shales using FDEM. Tatone and Grasselli[29] proposed a calibration procedure of microscopic parameters in two-dimensional FDEM. Lei et al.[30] implemented an empirical joint vonstitutive model into FDEM to analysis the behavior of fractured rocks.

Recently, Yan et al.[31–34] based on FDEM established two 2D/3D hydro-mechanical model (FDEM-flow2D/3D) to simulate fracturing driven by fluid. However, the method do not deal with problems of thermal cracking. Therefore, in this study we add a new feature to FDEM to simulate thermal cracking of rock, termed FDEM-TM. The organization of the paper is as follows. In Section 2, system equations and fracture criterion of joint element in FDEM are introduced briefly. Section 3 describes the coupled thermo-mechanical model, including thermal conduction model, thermal stress and the calculation processes of thermo-mechanical coupling, which is the core content of this paper. Finally, three examples with analytic solutions are given to verify the correctness of the coupled thermo-mechanical model to deal with problems of steady-state thermal conduction, unsteady-state thermal conduction and thermo-mechanical coupling, respectively. Moreover, a thermal cracking example is also given, and the simulation results of which are compared with experimental results.

2. Fundamentals of combined finite-discrete element method

In FDEM, the continuum in the problem domain is meshed into a finite element mesh composed by constant strain triangle (CST) elements, and joint elements with bonding effect are inserted in the common edge of adjacent triangular elements, as shown in Fig. 1. Thus, the triangular elements do not share nodes. Since the joint elements that connected two triangular elements may break and lost bonding effect, crack initiation, propagation and intersection can be simulated by the breakage of joint elements. While, before joint element broken, the continuum deformation can be represented by the triangular elements and joint elements with bonding effect.

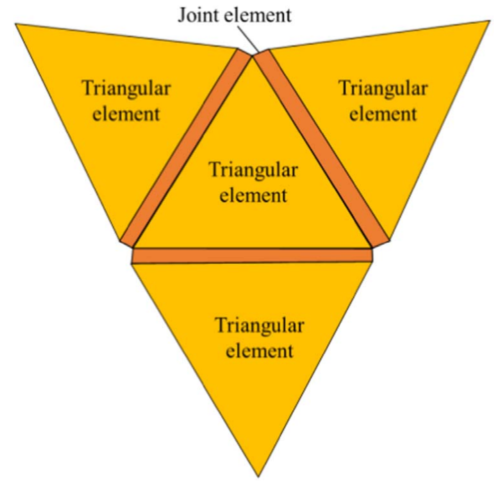

Fig. 1. Connection of triangular elements and joint elements.

2.1. System equations of FDEM

The dynamic equations of the system in FDEM are similar to that in discrete element method (DEM), given by

$$\mathbf{M}\ddot{x} + \mathbf{C}\dot{x} = \mathbf{F}(x) \tag{1}$$

where $\mathbf{M}$ and $\mathbf{C}$ are the mass matrix and damping matrix of nodes, respectively; $\mathbf{F}(x)$ represents unbalanced force vector of node, including the nodal force F_c caused by contact, the nodal force F_d by triangular element deformation, the nodal force F_e by the external load, and the nodal force F_j by bonding effect of joint element.

Since the statics problems are solved by dynamic relaxation method, the damping matrix $\mathbf{C}$ is used to consume kinetic energy of system. Damping matrix $\mathbf{C}$ is given by

$$\mathbf{C} = \mu\mathbf{I} \tag{2}$$

where μ is damping coefficient, $\mathbf{I}$ is unit matrix. According to the mass spring system of single degree of freedom, the critical damping coefficient μ_c is given by

$$\mu_c = 2h\sqrt{\rho E} \tag{3}$$

where h is the edge length of element, ρ is density, E is elastic modulus. If critical damping coefficient μ_c is used, kinetic energy of system can be consumed with the fastest speed.

According to Eq. (1), in each time step, coordinates and velocity of a node can be calculated by central difference method, as the following formula

$$\begin{aligned} v_i^{(t+\Delta t)} &= v_i^{(t)} + \sum F_i^{(t)}\frac{\Delta t}{m_n} \\ x_i^{(t+\Delta t)} &= x_i^{(t)} + v_i^{(t)}\Delta t \end{aligned} \tag{4}$$

where $F_i^{(t)}$, represents the total nodal force, Δt is time step, m_n is node mass, which is equal to one third of a triangular element's mass.

2.2. Fracture criterion of joint element in FDEM

Since crack initiation and propagation are simulated by the breakage of joint elements in FDEM, the fracture criterion of joint element is vital to cracking simulation. In this section we introduce the fracture

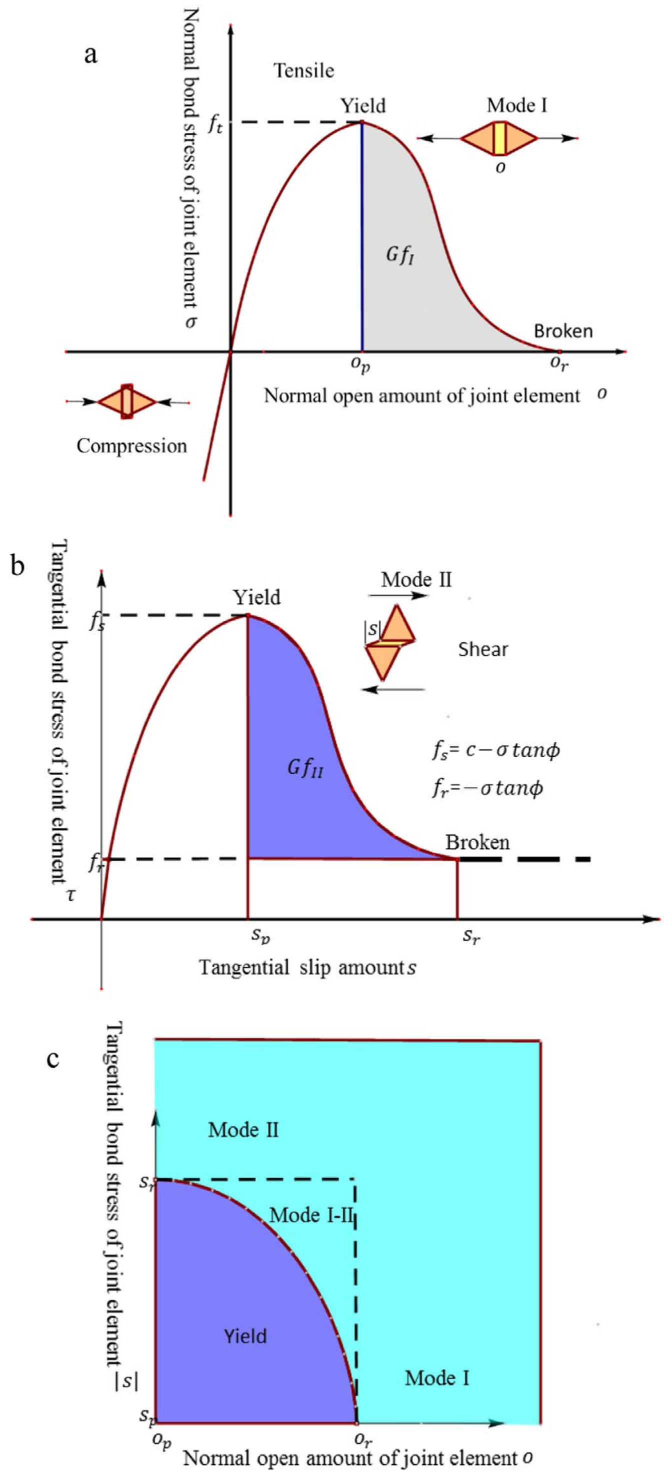

Fig. 2. Fracture criterion of a joint element in FEMDEM: (a) Relationship between normal bond stress and normal open amount; (b) relationship between tangential bond stress and tangential slip amount; (c) relationship between joint element yield, failure type and the normal opening amount and tangential slip amount.[36]

criterion of joint elements briefly.[35,36]

As shown in Fig. 2(a), when the normal opening amount of two sides of a joint element increases to a critical value o_{op}, the normal bond stress σ of the joint element is just equal to tensile strength f_t exactly. If the normal opening amount o of the joint element continues to increase, the normal bond stress σ gradually decreases until normal opening amount o is equal to the maximum open amount o_r. At this moment, the normal bond stress σ decreases to zero, and the joint element breaks, i.e. a tensile crack is generated.

As shown in Fig. 2(b), when the tangential slip amount of two sides of a joint element increases to a critical value s_p, the tangential bond stress τ of the joint element is just equal to shear strength f_s. If tangential slip amount s of the joint element continues to increase, the tangential bond stress τ gradually decreases until tangential slip amount s is equal to the maximum value s_r. At this moment, the tangential bond stress τ takes residual shear strength f_{sr}, and the joint element breaks, i.e. a shear crack is generated.

Shear strength f_s can be given by

$$f_s = c - \sigma \tan \phi \tag{5}$$

where c is cohesion, σ is the normal stress of joint element, because positive denotes tension, thus, in the front of σ a minus sign is added; ϕ is the internal friction angle.

Since c is equal to zero when a joint element breaking, the residual shear strength consists of friction force only and can be given by

$$f_{sr} = -\sigma \tan \phi \tag{6}$$

In addition to the above two fracture modes, there exists the third fracture mode, namely mixed I-II mode fracture (tensile shear mixing fracture), as shown in Fig. 2(c). In this fracture mode, two edges connected by a joint element is in open and slip mode simultaneously. Although the normal opening amount and tangential slip amount are less than the maximum normal opening amount o_r and maximum tangential slip amount s_r respectively, the normal opening amount and tangential slip amount satisfy the following condition

$$\left(\frac{o - o_p}{o_r - o_p}\right)^2 + \left(\frac{s - s_p}{s_r - s_p}\right)^2 \geq 1 \tag{7}$$

The maximum normal opening amount o_r depends on tensile strength f_t and energy release rate Gf_I of mode I crack. Similarly, tangential slip amount s_r depends on shear strength f_s and energy release rate Gf_{II} of mode II crack. The value of energy release rate Gf_I and Gf_{II} are area of the stained region in Fig. 2(a) and (b), respectively, which can be calculated as follows

$$Gf_I = \int_{o_p}^{o_r} \sigma(o)do \tag{8}$$

$$Gf_{II} = \int_{s_p}^{s_r} \tau(s)ds \tag{9}$$

3. The coupled thermo-mechanical model

Temperature changes in rock will cause thermal stress, and the stress change causes temperature change in turn. Since generally the effect of the stress changes on the temperature changes is considered relatively less, here we only consider thermal stress caused by the temperature in the coupled thermo-mechanical model.

3.1. Heat conduction model

According to Fourier's law of heat transfer, the heat flow rate of unit cross-sectional area in i direction can be expressed as

$$q_i = -k_{i,j}\frac{\partial T}{\partial x_j} \tag{10}$$

where q_i is thermal velocity along i direction, $k_{i,j}$ is thermal conductivity tensor, T is the temperature.

For any given mass M, the partial derivative of temperature with respect to time can be written by

$$\frac{\partial T}{\partial t} = \frac{Q_{net}}{C_p M} \tag{11}$$

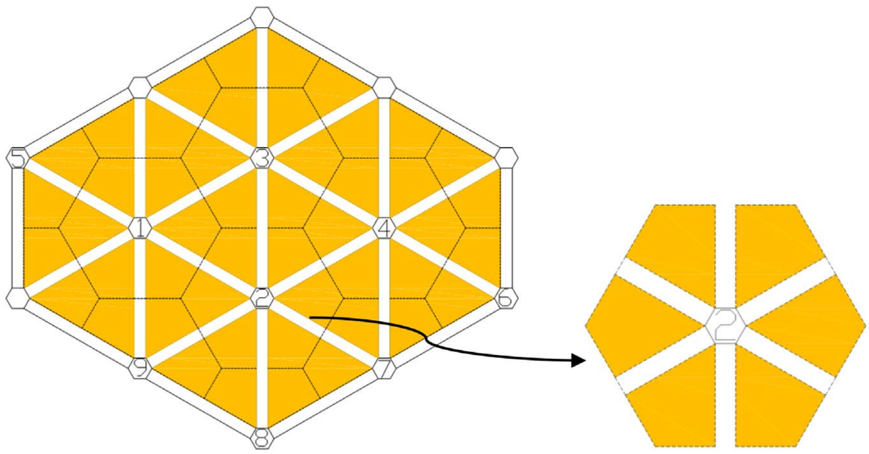

Fig. 3. Heat conduction representation model in FDEM-TM method.

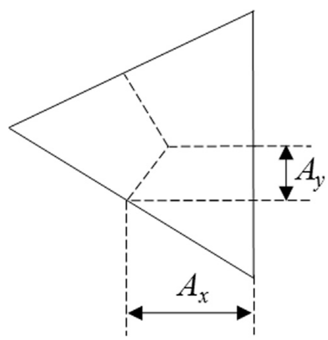

Fig. 4. Heat flow flowing into node 2.

where Q_{net} is the net heat into mass M at unit time, C_p is the heat capacity, and M is the mass.

Using the mesh in FDEM as shown in Fig. 1, we build a heat conduction model as shown in Fig. 3. In this model, the temperatures of hexagon area surrounding the node is represent by the temperature of nodes. We use the temperature of these discrete nodes (such as the circled nodes 1–9 in Fig. 3) to characterize the temperature field in the continuum. Here, we will describe how to calculate the temperature field in the continuum based on the topology as shown in Fig. 3.

In Fig. 3, we use node 2 as an example, where six triangular elements (Δ243 , Δ231 , Δ219 , Δ298 , Δ287 , Δ274) are connected with node 2. The hexagon area surrounding node 2 is called a domain, whose temperature is represented by temperature of node 2. Since the temperature of nodes 3, 4, 7, 8, 9 that connect directly with node 2 may be different from temperature of node 2, heat conduction may occur between these nodes. We use Δ231 (one of the triangular elements that connect to node 2) as an example. The temperature at the three nodes of the triangular element Δ231 are assumed as T_i, coordinates of (x_i, y_i) (i is the node number, i=1,2,3). Suppose the temperature within a triangular element obey linear distribution, the temperature of any point inside the triangular element can be expressed as

$$T = N_1T_1 + N_2T_2 + N_3T_3 \tag{12}$$

where $N_i = \frac{1}{2A}(a_i + b_ix + c_iy)$, which is interpolation shape functions of node and

$$a_i = \begin{vmatrix} x_j & y_j \\ x_m & y_m \end{vmatrix} = x_jy_m - x_my_j$$

$$b_i = -\begin{vmatrix} 1 & y_j \\ 1 & y_m \end{vmatrix} = y_j - y_m$$

$$c_i = -\begin{vmatrix} 1 & x_j \\ 1 & x_m \end{vmatrix} = -x_j + x_m$$

A is area of the triangular element.

Then, the temperature gradient in the triangular element can be expressed as follows

$$\begin{aligned} \frac{\partial T}{\partial x} &= \frac{\partial(N_1T_1 + N_2T_2 + N_3T_3)}{\partial x} = \frac{b_1T_1 + b_2T_2 + b_3T_3}{2A} \\ \frac{\partial T}{\partial y} &= \frac{\partial(N_1T_1 + N_2T_2 + N_3T_3)}{\partial y} = \frac{c_1T_1 + c_2T_2 + c_3T_3}{2A} \end{aligned} \tag{13}$$

Substituting Eq. (13) into to Eq. (10), we can obtain the heat flow rate along the x and y direction, as follows

$$\begin{aligned} q_x &= \frac{-(k_{11}(b_1T_1 + b_2T_2 + b_3T_3) + k_{12}(c_1T_1 + c_2T_2 + c_3T_3))}{2A} \\ q_y &= \frac{-(k_{21}(c_1T_1 + c_2T_2 + c_3T_3) + k_{22}(b_1T_1 + b_2T_2 + b_3T_3))}{2A} \end{aligned} \tag{14}$$

According to Fig. 4, per unit time, the heat flowing into node 2 can be calculated by the following formula[37]

$$Q = Q_x + Q_y = A_yq_x + A_xq_y \tag{15}$$

where A_x, A_y are coordinate difference between the midpoints of two edges connected by node 2.

Thus, we obtain the heat flowing into node 2 from the triangular element Δ231 per unit time. Similarly, the heat flowing into node 2 per unit time from other triangular elements that connect directly with node 2 can also be obtained. Ultimately, per unit time, the total heat flowing into node 2 can be represented as

$$Q_{total} = Q_{\Delta 243} + Q_{\Delta 231} + Q_{\Delta 219} + Q_{\Delta 298} + Q_{\Delta 287} + Q_{\Delta 274} \tag{16}$$

According to Eqs. (11) and (16), at next time step $(t + \Delta t)$, the temperature of node 2 is given by

$$T_2^{t+\Delta t} = T_2^t + \frac{Q_{total}}{C_pM}\Delta t \tag{17}$$

The temperature of other discrete nodes at next time step can be calculated similarly. Accordingly, we can obtain temperature field evolution of the system.

3.2. Thermal stress

As mentioned earlier, temperature change may cause thermal stress. The thermal stress can be calculated by

$$\Delta\sigma_{ij} = -\delta_{ij}3K^*\alpha\Delta T \tag{18}$$

where $\Delta\sigma_{ij}$ is the thermal stress increment, for plane stress problem $K^* = 6KG/(3K + 4G)$, K is the bulk modulus, G is the shear modulus, and α is thermal expansion coefficient.

Then, the thermal stress is applied to triangular elements as a body force. The equivalent nodal force of a triangular element caused by thermal stress can be given by

$$\mathbf{f}_n = -\delta_{ij} 3K^* \alpha \Delta T n_j L \tag{19}$$

where n_j is the outer normal vector of an edge of the triangle element, L is the edge length of the triangular element. To this point, the thermal stress induced by temperature change is considered in FDEM.

3.3. Implementation

From Sections 3.1 and 3.2, we can conclude that the coupled model can be implemented by the follow three main steps. First, the temperature distribution of the system is analyzed based on heat conduction equation. Second, the thermal stress caused by temperature is obtained and applied to triangular elements as a body load. Finally, mechanics fracture calculation is performed. On the basis of the above three steps, we can model and analyze the problem of rock fracture driven by temperature.

4. Examples

4.1. Steady-state heat conduction

A thick-walled cylinder[37] with inner diameter r_0=0.5 m and outer diameter R_0=5 m, the cylinder is assumed homogeneous isotropic media. The temperature at the inner boundary and outer boundary is kept constant T_{r0} and T_{R0}, respectively. The model is meshed into 939 triangular elements with average size of 0.5 m. We study the temperature distribution in the cylinder using the coupled thermo-mechanical model.

Analytical solution for the temperature distribution in the thick-walled cylinder is given by[6]

$$T = \frac{\ln(\frac{R_0}{r})}{\ln(\frac{R_0}{r_0})} T_{r_0} + \frac{\ln(\frac{r_0}{r})}{\ln(\frac{r_0}{R_0})} T_{R_0} \tag{20}$$

Let T_{R_0}=0 °C, T_{r_0}=100 °C. Since only the heat conduction is considered, the node coordinates of all elements are fixed regardless of the deformation. Calculation parameters include: heat capacity C_p=0.2 J/(kg °C), density ρ=2300 kg/m^3, thermal conductivity coefficient k=100 W/(m °C). A constant integration time step of 5×10^{-6} s is used. Final distribution of temperature field obtained by the coupled model is shown in Fig. 5.

Fig. 5(a) is the temperature distribution in the cylinder, the temperature gradually decrease from the inside to outside. The numerical solutions of the radial temperature distribution are is in good agreement with the analytical solution, as shown in Fig. 5(b), verifying the correctness of the coupled model in dealing with steady-state heat conduction.

4.2. Transient heat conduction

A rectangular plate[4] with length L and height H, the initial temperature in the plate is T_H, and the top and bottom boundary is adiabatic. The temperature at the right and left boundary is kept constant T_C. We study the temperature evolution of the plate with time. The mathematical description of the model is

$$\begin{aligned} &\frac{\partial T}{\partial t} = a\frac{\partial^2 T}{\partial x^2} && t>0,\ 0<x<l \\ &T|_{t=0} = T_0 && 0 \le x \le l \\ &T|_{x=0} = T|_{x=l} = 0 && t \ge 0 \end{aligned} \tag{21}$$

where $T_0 = T_H - T_c$. The analytic solution of Eq. (21) is given by[4]

$$T = \frac{4T_0}{\pi}\sum_{n=0}^{\infty}\frac{1}{(2n+1)}e^{-a(2n+1)^2\pi^2 t/l^2}\sin\frac{2n+1}{l}\pi x \tag{22}$$

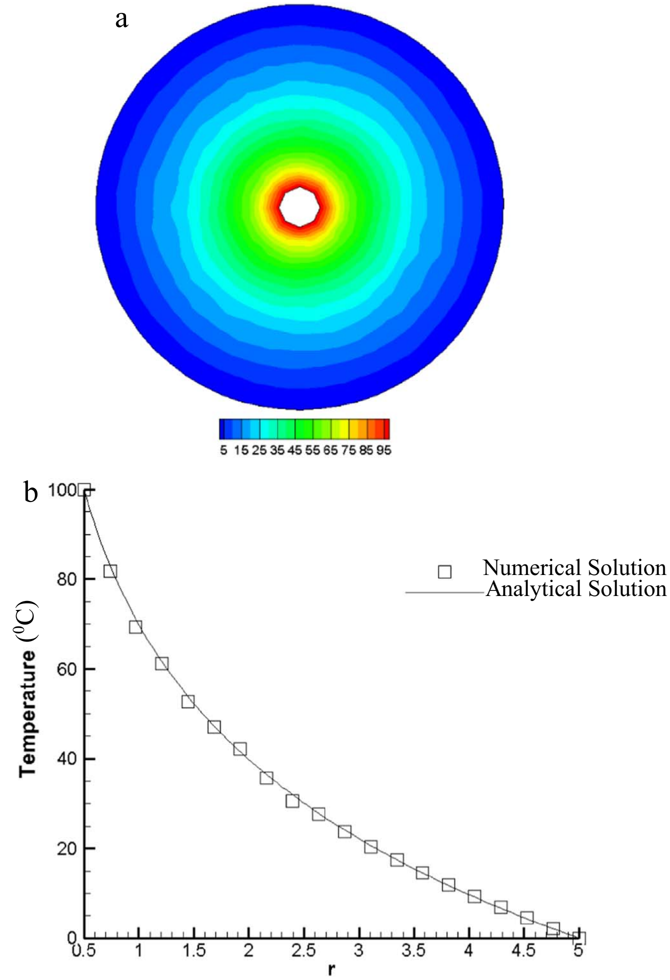

Fig. 5. Temperature distribution in the cylinder: (a) Temperature distribution in the cylinder; (b) numerical solutions and analytical solutions of the temperature distribution along radial.

Let L=2 m, H=0.5 m. All nodes coordinate are fixed without thermal deformation. The model is meshed into 280 triangular elements with average size of 0.5 m. The calculation parameters are the same as example in Section 4.1. A constant integration time step of 2.77×10^{-5} s is taken. We obtain temperature distribution in the plate, as shown in Fig. 6, using the couple model.

As shown in Fig. 6, the temperature of the rectangular plate gradually decreases at left and right, the temperature distribution is symmetrical about the middle line of the rectangular plate. Moreover, the decreasing zone of temperature expands from left and right sides to the middle of the plate. As shown in Fig. 7, the numerical solution agrees well with the analytic solution, demonstrating the correctness of the coupled model in dealing with transient heat conduction.

4.3. Deformation induced by temperature change

An unconstrained square plate with length L, the boundaries of the square plate is free, and the initial temperature of the square plate is T_0. Suppose the temperature of the square plate is changed to T_0+T, we calculate the strain of the plate induced by the temperature change. The analytical solution for thermally-induced strains can be expressed as follows[38]

$$\begin{cases} \Delta\varepsilon_{xx} = \Delta\varepsilon_{yy} = \begin{cases} \alpha\Delta T \\ \alpha(1+\nu)\Delta T \end{cases} \\ \Delta\varepsilon_{xy} = 0 \end{cases} \tag{23}$$

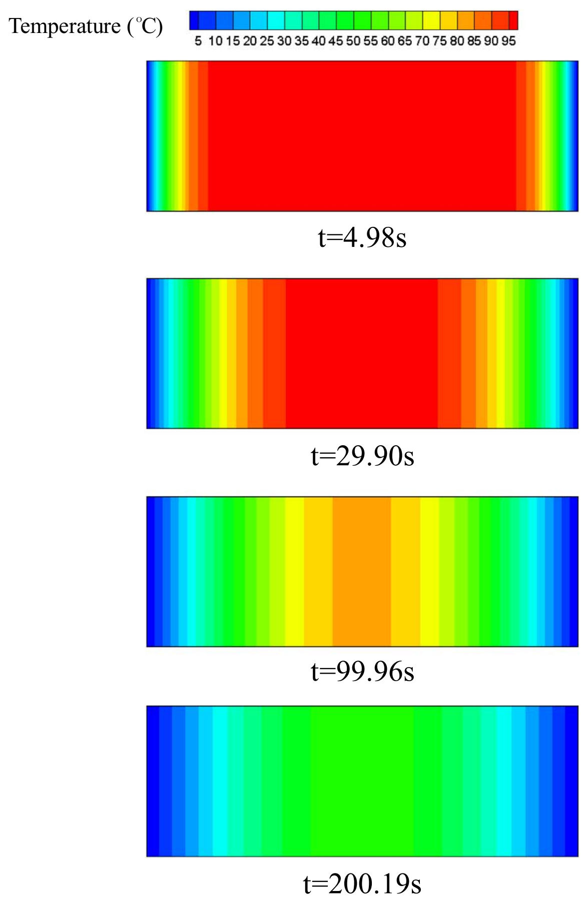

Fig. 6. The distribution of the temperature field at different time.

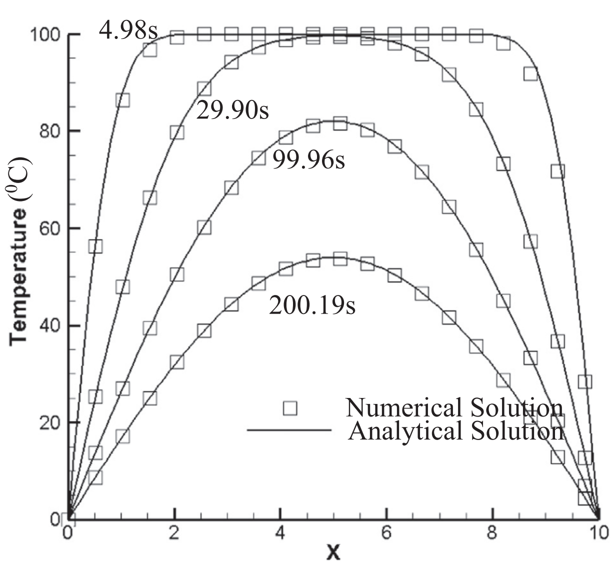

Fig. 7. Analytical solutions and numerical solutions of temperature distribution at different time.

where α is linear thermal expansion coefficient, ν is Poisson's ratio.

Let L=2 m. The model is meshed into 3294 triangular elements with average size of 0.05 m. Mechanical parameters and thermal parameters are listed in Table 1. A constant integration time step of 4.53×10^{-7} s is used. Two conditions are calculated with ΔT=100 °C and ΔT=−100 °C. The displacement vector obtained by the coupled model is as shown in Fig. 8.

As shown in Fig. 8(a), if the temperature increases, the square plate

Table 1
Parameters for example 3.

Parameters	Value
Rock	
Bulk density, ρ(kg/m^3)	2300
Young's modulus, E(GPa)	75.1
Poisson's ratio, υ	0.311
Joint element	
Tensile strength, f_t(MPa)	30
Internal cohesion, c(MPa)	50
Friction angle of intact material, ϕ (°)	35
Friction angle of fractures, ϕ (°)	35
Mode I fracture energy release rate, G_I(J/m^2)	2.0
Mode II fracture energy release rate, G_{II}(J/m^2)	10
Normal contact penalty, p_n(GPa)	300
Tangential contact penalty, p_t(GPa/m)	30
Fracture penalty, p_f(GPa)	300
Thermal parameters	
Thermal conductivity coefficient, k(W/(m.°C))	2.6
Heat capacity, C_p(J/(kg °C.s))	700
Thermal expansion coefficient α, (/°C)	3×10^{-6}

expands outward, and the closer a point is to the outer boundary, the greater it deforms. However, there is almost no displacement at the center of square plate. As shown in Fig. 8(b), if the temperature decreases, then square plate contracts inwardly, and the closer a point is to the outer boundary, the greater it deforms. Similarly, little no displacement at the center of the square plate. This indicates that the coupled model is reasonable from a qualitative point of view.

In addition, the strain of the square plate by the coupled model is: $\Delta\varepsilon_{xx}$=0.295111 and $\Delta\varepsilon_{yy}$=0.29511 if ΔT=100 °C, $\Delta\varepsilon_{xx}$=−0.295111 and $\Delta\varepsilon_{yy}$=−0.29511 if ΔT=−100 °C. According to Eq. (23), the analytic solution of the strain is $\Delta\varepsilon_{xx}=\Delta\varepsilon_{yy}=3\times10^{-4}$ if ΔT=100 °C and $\Delta\varepsilon_{xx}=\Delta\varepsilon_{yy}=-3\times10^{-4}$ if ΔT=−100 °C. The error of the numerical solutions is 1.6%. Numerical solutions are relatively close to analytical solutions, verifying the correctness of the coupled model in dealing with problems of thermo-mechanical coupling.

4.4. Thermal cracking example

The model consists of an outer cylinder and an inner disc. The radius of the cylinder and disc are 30 mm and 150 mm, respectively. Thermal expansion coefficient of the inner disc is 1.5 times as large as that of the outer cylinder. The initial temperature of the model is set at 25 °C, temperature of the model increase uniformly. Although the outer boundary of the model is free, thermal stress is observed in the model because the thermal expansion coefficient of the inner disc is different from that of the outer cylinder. This thermal stress is called thermal mismatch stress (TMS).[4] We study fracture morphology of the model using the coupled model and compare with experimental results. The model is meshed into 5620 triangular elements with average size of 0.005 m. Table 2 reports all mechanical and thermal parameters. A constant integration time step of 5×10^{-8} s is used. The simulation results are shown in Fig. 9.

As shown in Fig. 9, initially, the model cannot deform immediately under the effect of temperature. Thus, both the inner disc and the outer cylinder is in compression, as shown in step 1000 in Fig. 9. As temperature increases uniformly, the deformation of the inner disc is constrained by the outer cylinder. The deformation of the outer cylinder is less than that of the inner disc because the thermal expansion coefficient of outer cylinder is less than that of the inner disc. Therefore, the inner disc is in compression and the inner boundary of the outer cylinder in contacted with the inner disc is in tension. As shown from step 5000 to step 9000 in Fig. 9, the inner disc display dark blue, indicating the major principal stress is compression stress in the inner disc; however, the internal boundary of the outer cylinder displays deep red, indicating that the major principal stress at

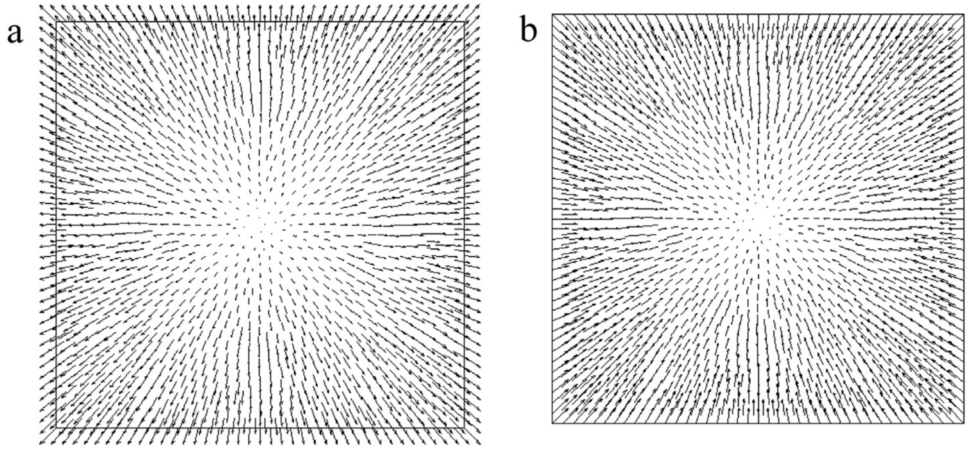

Fig. 8. Displacement vector diagram: (a) ΔT=100 °C; (b) ΔT=−100 °C.

Table 2
Parameters for example 4.

Parameters	Inner disc	Outer cylinder
Rock		
Bulk density, ρ(kg/m^3)	2300	2300
Young's modulus, E(GPa)	40	20
Poisson's ratio, υ	0.3	0.2
Joint element		
Tensile strength, f_t(MPa)	20	10
Internal cohesion, c(MPa)	40	20
Friction angle of intact material, ϕ (°)	30	30
Friction angle of fractures, ϕ (°)	30	30
Mode I fracture energy release rate, G_I(J/m^2)	2	2
Mode II fracture energy release rate, G_{II}(J/m^2)	10	10
Normal contact penalty, p_n(GPa)	40	40
Tangential contact penalty, p_t(GPa/m)	4	4
Fracture penalty, p_f(GPa)	40	40
Thermal parameters		
Thermal conductivity coefficient, k(W/(m.°C))	1.2	1.2
Heat capacity, C_p(J/(kg °C.s))	900	900
Thermal expansion coefficient α, (/°C)	1.5×10^{-6}	1×10^{-6}

the inner boundary of the cylinder is in tension. At the same time, it can be seen from step 5000 to step 9000 in Fig. 9, as temperature increases, the tensile stress (i.e. major principal stress) within the inner boundary of the cylinder increases. When the tensile stress exceeds the tensile strength of the outer cylinder, a tensile crack is generated in the outer cylinder. The tensile crack causes the release of tensile stress near the crack, the tensile stress concentration zone near the inner boundary of the cylinder disappears, while new tensile stress concentration is generated at the crack tip as shown in Fig. 9 in step 100,000. Crack initiation occurs in the bottom right corner of the inner boundary of the cylinder. After that, there also another crack at the top left of the inner boundary of the cylinder is generated. Since the model mesh is incomplete symmetry, the two cracks do not extend at the same time. But the final result is that the cracks at the top left and bottom right extend approximately symmetrical from the inside to the outside. Tensile stress concentration is generated at the crack tip, the zone of tensile stress concentration moves outward with the crack tip. However, the range of tensile stress concentration zone within the inner boundary (red color) gradually decreases because the tensile stress within the inner boundary of outer cylinder releases due to crack initiation. Although there are still two zones of tensile stress concentration within the inner boundary of the outer cylinder, the tensile stress in this region is less than tensile strength of the outer cylinder. Thus, no new cracks are generated at the inner boundary of the cylinder, as shown in steps 102,000–105,000 in Fig. 9. Finally, the outer cylinder is cut into two blocks approximately symmetrical.

As shown in Fig. 10(b), in the experiment, cracking initiates at the inner boundary of the outer cylinder firstly, and then it extends outward to form radial cracks. The propagation trend of the simulation results is consistent with the experimental results,[39] which verifies the correctness of the coupled thermo-mechanical model in simulating thermal cracking. Of course, since the heterogeneity of experimental material and mesh-dependence, the initiation position of both cannot be completely consistent.

As shown in Fig. 11, the maximum value of the major principal stress is in the inner boundary of the cylinder before crack initiation. When the maximum value of the major principal stress exceeds the tensile strength of the cylinder, cracking initiates. Then, the maximum value of the major principal stress is located within the inner boundary of cylinder and the crack tip. Since the crack initiation, the maximum principal stress at the inner boundary of the outer cylinder release, the maximum value of the maximum principal stress begins to decrease. From Fig. 11, the maximum value of the maximum major principal stress begins to decrease near step 95,000, combined with Fig. 9, at this moment, it is just located between step 90,000 (before crack initiation) to step 100,000 (cracks generated). Moreover, after crack initiation, the maximum value of the maximum major principal stress on the inner boundary is lower than the tensile strength, which inhibits new cracks from generating at the inner boundary of the cylinder. The results further validate the previous discussion: as temperature continues to increase, when tensile stress at the internal boundary of the outer cylinder is larger than the tensile strength, cracking is initiated. Crack generation leads to the stress release, which makes tensile stress on the inner boundary of cylinder decrease (low than the tensile strength of material) and no new cracks are generated in the inner boundary.

5. Conclusions

This study presents a coupled thermo-mechanical model (FDEM-TM) based on FDEM for simulating thermal cracking of rock. Three main parts are involved in the proposed coupled model: the thermal conductivity, thermal stress calculation module, and the mechanic fracture calculation module. Through three examples with analytic solution, we verify the correctness of the coupled model in dealing with problems of steady-state, unsteady-state heat conduction and thermo-mechanical coupling, respectively. Finally, a thermal cracking example is given and compared with experiment results verifying the correctness of the coupled model in dealing with the problem of thermal cracking.

Acknowledgements

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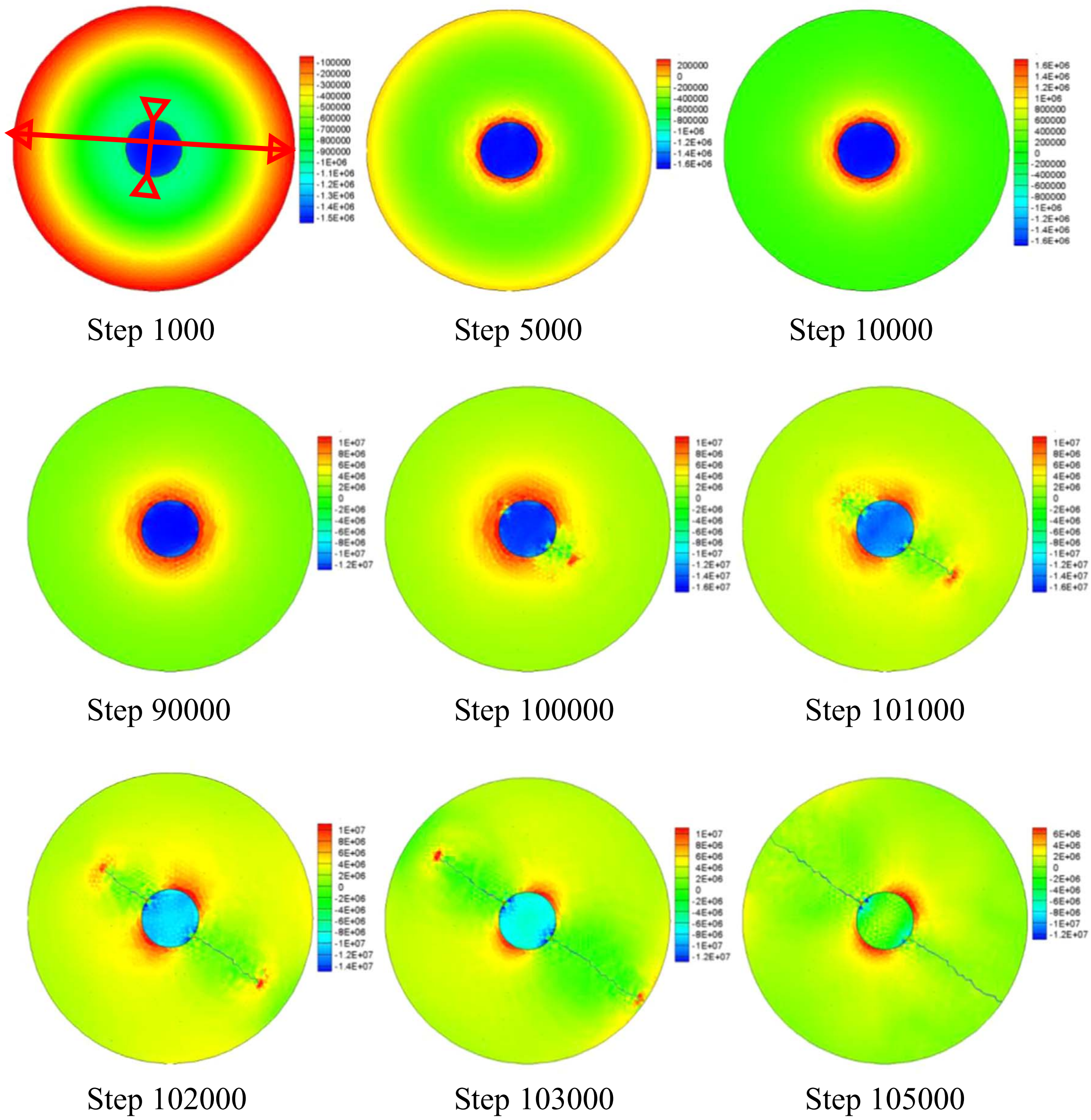

Fig. 9. Major principal stress distribution (Pa, positive donate tensile stress). (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

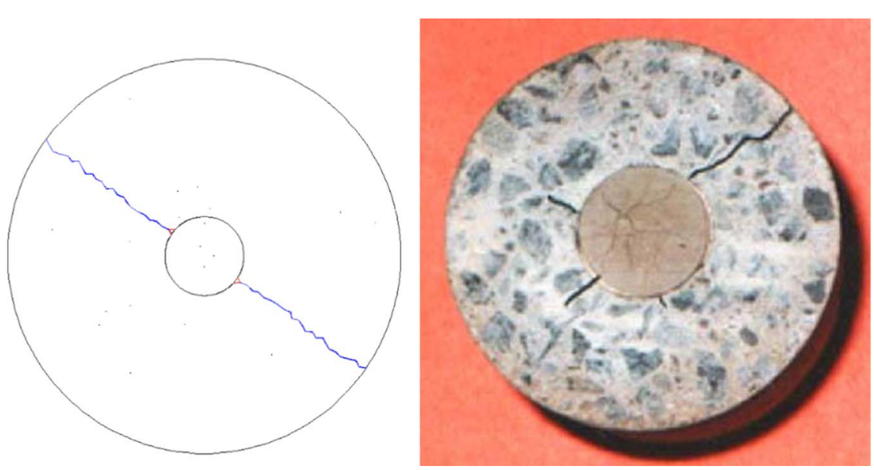

Fig. 10. Thermal cracking morphology, ultimately: (a) Thermal cracking morphology of simulation; (b) thermal cracking morphology of experimental results.[39]

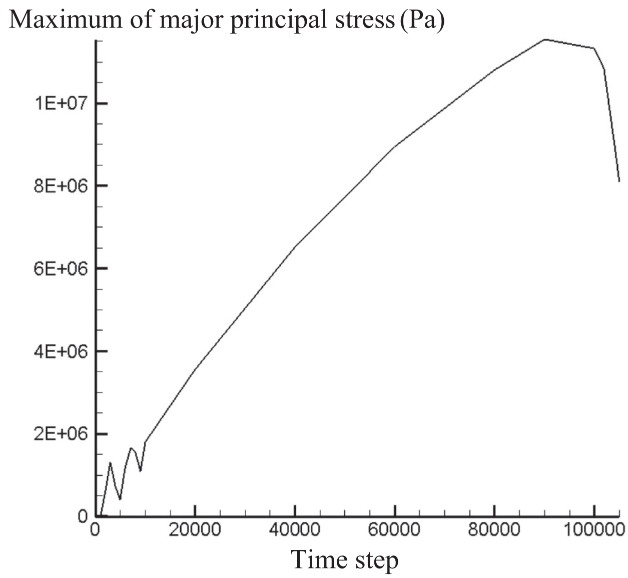

Fig. 11. Maximum of maximum principal stress with time step.

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